201. **Checking Account** John has $\$148$ in his checking account. He writes a check for $\$83$. What is the new balance in his checking account?

202. **Checking Account** Ellie has $\$426$ in her checking account. She writes a check for $\$152$. What is the new balance in her checking account?

203. **Checking Account** Gina has $\$210$ in her checking account. She writes a check for $\$250$. What is the new balance in her checking account?

204. **Checking Account** Frank has $\$94$ in his checking account. He writes a check for $\$110$. What is the new balance in his checking account?

205. **Checking Account** Bill has a balance of $-\$14$ in his checking account. He deposits $\$40$ to the account. What is the new balance?

206. **Checking Account** Patty has a balance of $-\$23$ in her checking account. She deposits $\$80$ to the account. What is the new balance?

Everyday Math

207. **Camping** Rene is on an Alpine hike. The temperature is $-7°$. Rene's sleeping bag is rated "comfortable to $-20°$". How much can the temperature change before it is too cold for Rene's sleeping bag?

208. **Scuba Diving** Shelly's scuba watch is guaranteed to be watertight to -100 feet. She is diving at -45 feet on the face of an underwater canyon. By how many feet can she change her depth before her watch is no longer guaranteed?

Writing Exercises

209. Explain why the difference of 9 and -6 is 15.

210. Why is the result of subtracting $3-(-4)$ the same as the result of adding $3+4$?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
model subtraction of integers.			
simplify expressions with integers.			
evaluate variable expressions with integers.			
translate word phrases to algebraic expressions.			
subtract integers in applications.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

3.4 Multiply and Divide Integers

Learning Objectives

By the end of this section, you will be able to:

- Multiply integers
- Divide integers
- Simplify expressions with integers
- Evaluate variable expressions with integers
- Translate word phrases to algebraic expressions

☑ **BE PREPARED** 3.7 Before you get started, take this readiness quiz.

Translate the quotient of 20 and 13 into an algebraic expression.
If you missed this problem, review Example 1.67.

☑ **BE PREPARED** 3.8 Add: $-5+(-5)+(-5)$.
If you missed this problem, review Example 3.21.

☑ **BE PREPARED** 3.9 Evaluate $n + 4$ when $n = -7$.
If you missed this problem, review Example 3.23.

Multiply Integers

Since multiplication is mathematical shorthand for repeated addition, our counter model can easily be applied to show multiplication of integers. Let's look at this concrete model to see what patterns we notice. We will use the same examples that we used for addition and subtraction.

We remember that $a \cdot b$ means add a, b times. Here, we are using the model shown in Figure 3.19 just to help us discover the pattern.

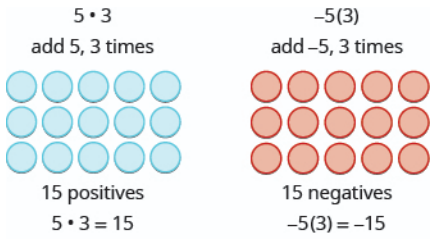


Figure 3.19

Now consider what it means to multiply 5 by -3. It means subtract 5, 3 times. Looking at subtraction as *taking away*, it means to take away 5, 3 times. But there is nothing to take away, so we start by adding neutral pairs as shown in Figure 3.20.

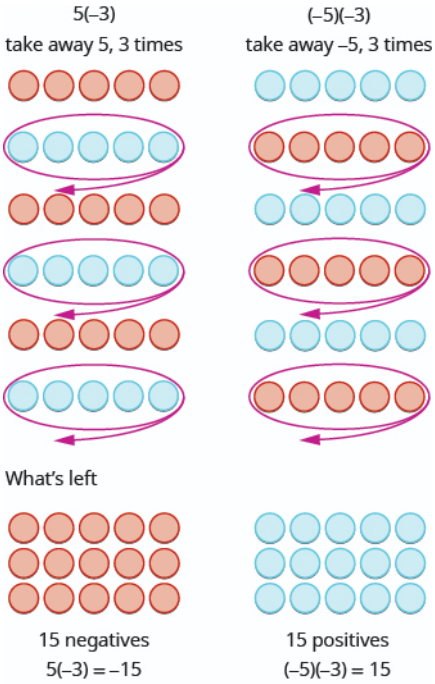


Figure 3.20

In both cases, we started with $\mathbf{15}$ neutral pairs. In the case on the left, we took away $\mathbf{5}, \mathbf{3}$ times and the result was $\mathbf{-15}$. To multiply $(-5)(-3)$, we took away $\mathbf{-5}, \mathbf{3}$ times and the result was $\mathbf{15}$. So we found that

$$5 \cdot 3 = 15 \qquad -5(3) = -15$$
$$5(-3) = -15 \qquad (-5)(-3) = 15$$

Notice that for multiplication of two signed numbers, when the signs are the same, the product is positive, and when the signs are different, the product is negative.

Multiplication of Signed Numbers

The sign of the product of two numbers depends on their signs.

Same signs	Product
•Two positives •Two negatives	Positive Positive

Different signs	Product
•Positive • negative •Negative • positive	Negative Negative

EXAMPLE 3.47

Multiply each of the following:

ⓐ $-9 \cdot 3$ ⓑ $-2(-5)$ ⓒ $4(-8)$ ⓓ $7 \cdot 6$

Solution

ⓐ

	$-9 \cdot 3$
Multiply, noting that the signs are different and so the product is negative.	-27

Table 3.5

ⓑ

	$-2(-5)$
Multiply, noting that the signs are the same and so the product is positive.	10

Table 3.6

ⓒ

	$4(-8)$
Multiply, noting that the signs are different and so the product is negative.	-32

Table 3.7

ⓓ	
	$7 \cdot 6$
The signs are the same, so the product is positive.	42

Table 3.8

TRY IT 3.93 Multiply:

ⓐ $-6 \cdot 8$ ⓑ $-4(-7)$ ⓒ $9(-7)$ ⓓ $5 \cdot 12$

TRY IT 3.94 Multiply:

1. ⓐ $-8 \cdot 7$
2. ⓑ $-6(-9)$
3. ⓒ $7(-4)$
4. ⓓ $3 \cdot 13$

When we multiply a number by 1, the result is the same number. What happens when we multiply a number by -1? Let's multiply a positive number and then a negative number by -1 to see what we get.

$-1 \cdot 4$	$-1(-3)$
-4	3
-4 is the opposite of $\mathbf{4}$	$\mathbf{3}$ is the opposite of -3

Each time we multiply a number by -1, we get its opposite.

Multiplication by -1

Multiplying a number by -1 gives its opposite.

$$-1a = -a$$

EXAMPLE 3.48

Multiply each of the following:

ⓐ $-1 \cdot 7$ ⓑ $-1(-11)$

Solution

ⓐ	
The signs are different, so the product will be negative.	$-1 \cdot 7$
Notice that −7 is the opposite of 7.	-7

ⓑ

The signs are the same, so the product will be positive.	$-1(-11)$
Notice that 11 is the opposite of −11.	11

TRY IT 3.95 Multiply.

ⓐ $-1 \cdot 9$ ⓑ $-1 \cdot (-17)$

TRY IT 3.96 Multiply.

ⓐ $-1 \cdot 8$ ⓑ $-1 \cdot (-16)$

Divide Integers

Division is the inverse operation of multiplication. So, $15 \div 3 = 5$ because $5 \cdot 3 = 15$ In words, this expression says that **15** can be divided into **3** groups of **5** each because adding five three times gives **15**. If we look at some examples of multiplying integers, we might figure out the rules for dividing integers.

$$5 \cdot 3 = 15 \text{ so } 15 \div 3 = 5 \qquad -5(3) = -15 \text{ so } -15 \div 3 = -5$$
$$(-5)(-3) = 15 \text{ so } 15 \div (-3) = -5 \qquad 5(-3) = -15 \text{ so } -15 \div -3 = 5$$

Division of signed numbers follows the same rules as multiplication. When the signs are the same, the quotient is positive, and when the signs are different, the quotient is negative.

Division of Signed Numbers

The sign of the quotient of two numbers depends on their signs.

Same signs	Quotient
•Two positives •Two negatives	Positive Positive

Different signs	Quotient
•Positive & negative •Negative & positive	Negative Negative

Remember, you can always check the answer to a division problem by multiplying.

EXAMPLE 3.49

Divide each of the following:

ⓐ $-27 \div 3$ ⓑ $-100 \div (-4)$

Solution

ⓐ

	$-27 \div 3$
Divide, noting that the signs are different and so the quotient is negative.	-9

Table 3.9

ⓑ

	$-100 \div (-4)$
Divide, noting that the signs are the same and so the quotient is positive.	25

Table 3.10

TRY IT 3.97 Divide:

ⓐ $-42 \div 6$ ⓑ $-117 \div (-3)$

TRY IT 3.98 Divide:

ⓐ $-63 \div 7$ ⓑ $-115 \div (-5)$

Just as we saw with multiplication, when we divide a number by 1, the result is the same number. What happens when we divide a number by -1? Let's divide a positive number and then a negative number by -1 to see what we get.

$8 \div (-1)$	$-9 \div (-1)$
-8	9
-8 is the opposite of 8	9 is the opposite of -9

When we divide a number by, -1 we get its opposite.

Division by -1

Dividing a number by -1 gives its opposite.

$$a \div (-1) = -a$$

EXAMPLE 3.50

Divide each of the following:

ⓐ $16 \div (-1)$ ⓑ $-20 \div (-1)$

Solution

ⓐ

	$16 \div (-1)$
The dividend, 16, is being divided by –1.	-16

Dividing a number by –1 gives its opposite.	
Notice that the signs were different, so the result was negative.	

Table 3.11

ⓑ	
	$-20 \div (-1)$
The dividend, –20, is being divided by –1.	20
Dividing a number by –1 gives its opposite.	

Table 3.12

Notice that the signs were the same, so the quotient was positive.

TRY IT 3.99 Divide:

ⓐ $6 \div (-1)$ ⓑ $-36 \div (-1)$

TRY IT 3.100 Divide:

ⓐ $28 \div (-1)$ ⓑ $-52 \div (-1)$

Simplify Expressions with Integers

Now we'll simplify expressions that use all four operations–addition, subtraction, multiplication, and division–with integers. Remember to follow the order of operations.

EXAMPLE 3.51

Simplify: $7(-2) + 4(-7) - 6$.

Solution

We use the order of operations. Multiply first and then add and subtract from left to right.

	$7(-2) + 4(-7) - 6$
Multiply first.	$-14 + (-28) - 6$
Add.	$-42 - 6$
Subtract.	-48

TRY IT 3.101 Simplify:

$8(-3) + 5(-7) - 4$

TRY IT 3.102 Simplify:

$$9(-3)+7(-8)-1$$

EXAMPLE 3.52

Simplify:

ⓐ $(-2)^4$ ⓑ -2^4

Solution

The exponent tells how many times to multiply the base.

ⓐ The exponent is 4 and the base is -2. We raise -2 to the fourth power.

	$(-2)^4$
Write in expanded form.	$(-2)(-2)(-2)(-2)$
Multiply.	$4(-2)(-2)$
Multiply.	$-8(-2)$
Multiply.	16

ⓑ The exponent is 4 and the base is 2. We raise 2 to the fourth power and then take the opposite.

	-2^4
Write in expanded form.	$-(2\cdot 2\cdot 2\cdot 2)$
Multiply.	$-(4\cdot 2\cdot 2)$
Multiply.	$-(8\cdot 2)$
Multiply.	-16

TRY IT 3.103 Simplify:

ⓐ $(-3)^4$ ⓑ -3^4

TRY IT 3.104 Simplify:

ⓐ $(-7)^2$ ⓑ -7^2

EXAMPLE 3.53

Simplify: $12-3(9-12)$.

Solution

According to the order of operations, we simplify inside parentheses first. Then we will multiply and finally we will subtract.

	$12-3(9-12)$
Subtract the parentheses first.	$12-3(-3)$
Multiply.	$12-(-9)$
Subtract.	21

TRY IT 3.105 Simplify:

$17-4(8-11)$

TRY IT 3.106 Simplify:

$16-6(7-13)$

EXAMPLE 3.54

Simplify: $8(-9)\div(-2)^3$.

Solution

We simplify the exponent first, then multiply and divide.

	$8(-9)\div(-2)^3$
Simplify the exponent.	$8(-9)\div(-8)$
Multiply.	$-72\div(-8)$
Divide.	9

TRY IT 3.107 Simplify:

$12(-9)\div(-3)^3$

TRY IT 3.108 Simplify:

$18(-4)\div(-2)^3$

EXAMPLE 3.55

Simplify: $-30\div 2+(-3)(-7)$.

Solution

First we will multiply and divide from left to right. Then we will add.

	$-30 \div 2 + (-3)(-7)$
Divide.	$-15 + (-3)(-7)$
Multiply.	$-15 + 21$
Add.	6

TRY IT 3.109 Simplify:

$-27 \div 3 + (-5)(-6)$

TRY IT 3.110 Simplify:

$-32 \div 4 + (-2)(-7)$

Evaluate Variable Expressions with Integers

Now we can evaluate expressions that include multiplication and division with integers. Remember that to evaluate an expression, substitute the numbers in place of the variables, and then simplify.

EXAMPLE 3.56

Evaluate $2x^2 - 3x + 8$ when $x = -4$.

Solution

	$2x^2 - 3x + 8$
Substitute -4 for x.	$2(-4)^2 - 3(-4) + 8$
Simplify exponents.	$2(16) - 3(-4) + 8$
Multiply.	$32 - (-12) + 8$
Subtract.	$44 + 8$
Add.	52

Keep in mind that when we substitute -4 for x, we use parentheses to show the multiplication. Without parentheses, it would look like $2 \cdot -4^2 - 3 \cdot -4 + 8$.

TRY IT 3.111 Evaluate:

$3x^2 - 2x + 6$ when $x = -3$

TRY IT 3.112 Evaluate:

$4x^2 - x - 5$ when $x = -2$

EXAMPLE 3.57

Evaluate $3x + 4y - 6$ when $x = -1$ and $y = 2$.

Solution

	$3x + 4y - 6$
Substitute $x = -1$ and $y = 2$.	$3(-1) + 4(2) - 6$
Multiply.	$-3 + 8 - 6$
Simplify.	-1

TRY IT 3.113 Evaluate:

$7x + 6y - 12$ when $x = -2$ and $y = 3$

TRY IT 3.114 Evaluate:

$8x - 6y + 13$ when $x = -3$ and $y = -5$

Translate Word Phrases to Algebraic Expressions

Once again, all our prior work translating words to algebra transfers to phrases that include both multiplying and dividing integers. Remember that the key word for multiplication is *product* and for division is *quotient*.

EXAMPLE 3.58

Translate to an algebraic expression and simplify if possible: the product of -2 and 14.

Solution

The word *product* tells us to multiply.

	the product of -2 and 14
Translate.	$(-2)(14)$
Simplify.	-28

TRY IT 3.115 Translate to an algebraic expression and simplify if possible:

the product of -5 and 12

TRY IT 3.116 Translate to an algebraic expression and simplify if possible:

the product of 8 and -13

EXAMPLE 3.59

Translate to an algebraic expression and simplify if possible: the quotient of -56 and -7.

Solution

The word *quotient* tells us to divide.

	the quotient of −56 and −7
Translate.	$-56 \div (-7)$
Simplify.	8

TRY IT 3.117 Translate to an algebraic expression and simplify if possible:

the quotient of -63 and -9

TRY IT 3.118 Translate to an algebraic expression and simplify if possible:

the quotient of -72 and -9

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Multiplying Integers Using Color Counters (https://openstaxcollege.org/l/24Multiplyinteg)
Multiplying Integers Using Color Counters With Neutral Pairs (https://openstaxcollege.org/l/24Multiplyneutr)
Multiplying Integers Basics (https://openstaxcollege.org/l/24Multiplybasic)
Dividing Integers Basics (https://openstaxcollege.org/l/24Dividebasic)
Ex. Dividing Integers (https://openstaxcollege.org/l/24Divideinteger)
Multiplying and Dividing Signed Numbers (https://openstaxcollege.org/l/24Multidivisign)

SECTION 3.4 EXERCISES

Practice Makes Perfect

Multiply Integers

In the following exercises, multiply each pair of integers.

211. $-4 \cdot 8$
212. $-3 \cdot 9$
213. $-5(7)$
214. $-8(6)$
215. $-18(-2)$
216. $-10(-6)$
217. $9(-7)$
218. $13(-5)$
219. $-1 \cdot 6$
220. $-1 \cdot 3$
221. $-1(-14)$
222. $-1(-19)$

Divide Integers

In the following exercises, divide.

223. $-24 \div 6$
224. $-28 \div 7$
225. $56 \div (-7)$
226. $35 \div (-7)$
227. $-52 \div (-4)$
228. $-84 \div (-6)$
229. $-180 \div 15$
230. $-192 \div 12$
231. $49 \div (-1)$
232. $62 \div (-1)$

Simplify Expressions with Integers

In the following exercises, simplify each expression.

233. $5(-6)+7(-2)-3$

234. $8(-4)+5(-4)-6$

235. $-8(-2)-3(-9)$

236. $-7(-4)-5(-3)$

237. $(-5)^3$

238. $(-4)^3$

239. $(-2)^6$

240. $(-3)^5$

241. -4^2

242. -6^2

243. $-3(-5)(6)$

244. $-4(-6)(3)$

245. $-4 \cdot 2 \cdot 11$

246. $-5 \cdot 3 \cdot 10$

247. $(8-11)(9-12)$

248. $(6-11)(8-13)$

249. $26-3(2-7)$

250. $23-2(4-6)$

251. $-10(-4) \div (-8)$

252. $-8(-6) \div (-4)$

253. $65 \div (-5)+(-28) \div (-7)$

254. $52 \div (-4)+(-32) \div (-8)$

255. $9-2[3-8(-2)]$

256. $11-3[7-4(-2)]$

257. $(-3)^2-24 \div (8-2)$

258. $(-4)^2-32 \div (12-4)$

Evaluate Variable Expressions with Integers

In the following exercises, evaluate each expression.

259. $-2x+17$ when
ⓐ $x=8$ ⓑ $x=-8$

260. $-5y+14$ when
ⓐ $y=9$ ⓑ $y=-9$

261. $10-3m$ when
ⓐ $m=5$
ⓑ $m=-5$

262. $18-4n$ when
ⓐ $n=3$ ⓑ $n=-3$

263. p^2-5p+5 when $p=-1$

264. q^2-2q+9 when $q=-2$

265. $2w^2-3w+7$ when $w=-2$

266. $3u^2-4u+5$ when $u=-3$

267. $6x-5y+15$ when $x=3$ and $y=-1$

268. $3p-2q+9$ when $p=8$ and $q=-2$

269. $9a-2b-8$ when $a=-6$ and $b=-3$

270. $7m-4n-2$ when $m=-4$ and $n=-9$

Translate Word Phrases to Algebraic Expressions

In the following exercises, translate to an algebraic expression and simplify if possible.

271. The product of -3 and 15

272. The product of -4 and 16

273. The quotient of -60 and -20

274. The quotient of -40 and -20

275. The quotient of -6 and the sum of a and b

276. The quotient of -7 and the sum of m and n

277. The product of -10 and the difference of p and q

278. The product of -13 and the difference of c and d

Everyday Math

279. **Stock market** Javier owns 300 shares of stock in one company. On Tuesday, the stock price dropped \$12 per share. What was the total effect on Javier's portfolio?

280. **Weight loss** In the first week of a diet program, eight women lost an average of 3 pounds each. What was the total weight change for the eight women?

Writing Exercises

281. In your own words, state the rules for multiplying two integers.

282. In your own words, state the rules for dividing two integers.